Chapter 6

OPERATIONAL AMPLIFIERS

CHAPTER OUTLINE

6-1 Introduction to Operational Amplifiers

6-2 Op-Amp parameters

6-3 Negative Feedback in Op-Amps

6-4 Op-Amp Configurations

6-5 Troubleshooting

Discrete components (Diode, BJT, FET,…)  Integrated circuit (IC)

KEY OBJECTIVES

After completing this chapter, you should be able to

- Describe the basic op-amp and its **characteristics**
- Discuss several op-amp **parameters**
- Explain **negative feedback** in op-amp circuits
- Analyze three op-amp **configurations** and explain the closed-loop **frequency response** of an op-amp
- Troubleshoot op-amp circuits

SCIENCE HIGHLIGHT

- **Model**
  - Models in science are useful for predicting the results of a new problem.
  - Goal of a model
    - ✓ To describe laws of physics with simplicity
    - ✓ To predict the new result with reasonable accuracy
  - Example
    - ✓ $I = V/R, \quad \oiint D \cdot dS = \sum q, \quad \oint E \cdot dl = -\frac{d}{dt} \oiint B \cdot dS$
    - ✓ BJT: current-controlled current source
    - ✓ FET: voltage-controlled current source
    - ✓ $F = ma$

SCIENCE HIGHLIGHT

- **Model –** predict the result with reasonable accuracy
  - Suitable conditions
    - ✓ BJT: linear current-controlled current source
    - ✓ FET: linear voltage-controlled current source

    } (Small signals)

    - ✓ $F = ma$ (Low speed)
  - Circuit model
    - ✓ To predict the circuit's behavior accurately
  - Ideal model
    - ✓ To simplify the circuit
    - ✓ Leave out a lot of extraneous details
    - ✓ Excellent for understanding basic behavior

6-1 Introduction to Operational Amplifiers

- ✓ Developed in the I940s,
  - -- used primarily to perform mathematical operations
  - -- constructed with vacuum tubes
  - -- worked with high voltages.
- ✓ Today
  - -- are linear integrated circuits
  - -- use relatively low supply voltages
  - -- are reliable and inexpensive.

6-1-1 Symbol and Terminals

6-1-2 The Ideal Op-Amp

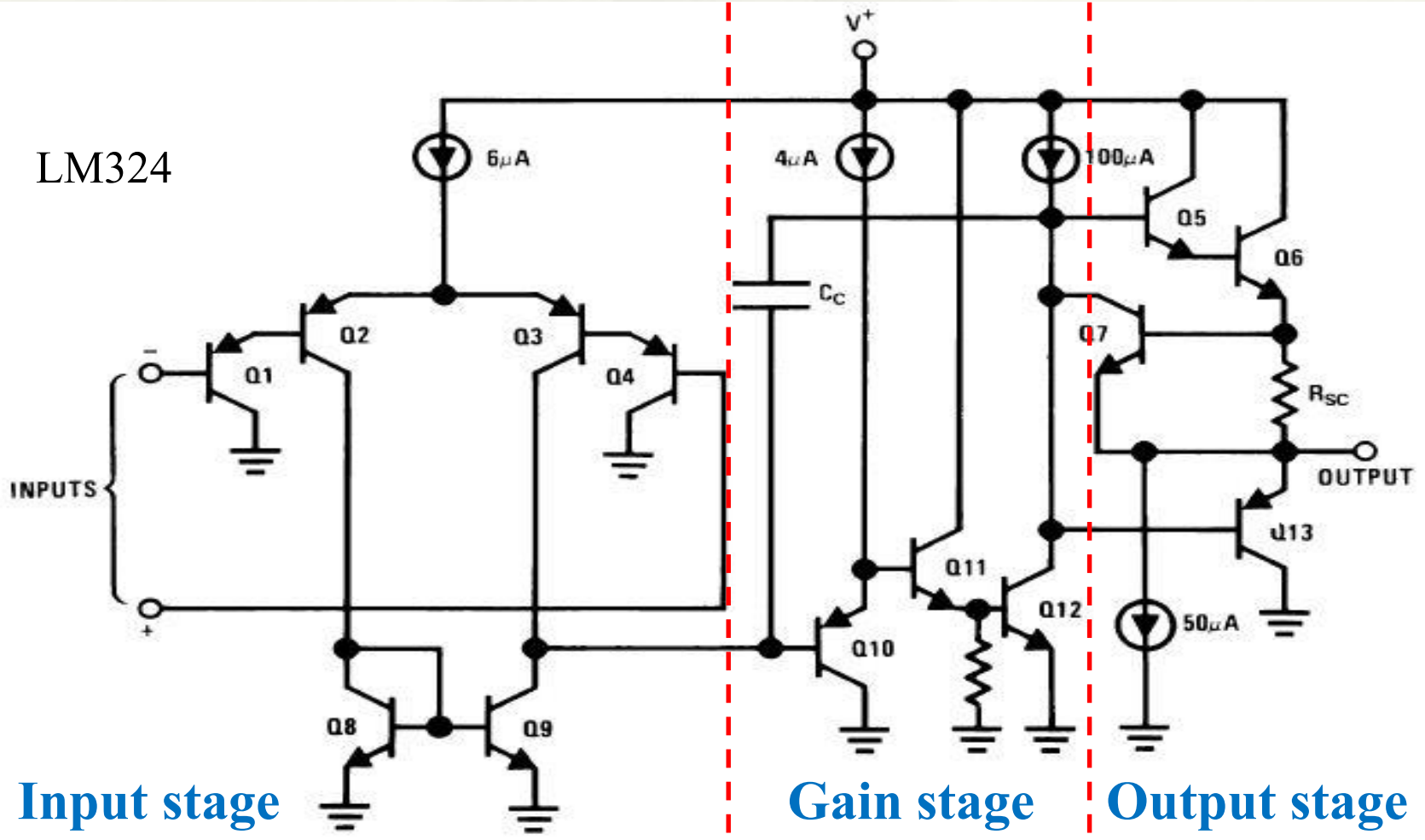
6-1-3 The Practical Op-Amp

6-1 Introduction to Operational Amplifiers

- **Operational amplifier**
  - An operational amplifier (op-amp) is an electronic device that amplifies the difference voltage between the two inputs.
  - Op-amps were used primarily to perform mathematical operations such as addition, subtraction, integration, and differentiation, thus the term *operational*.

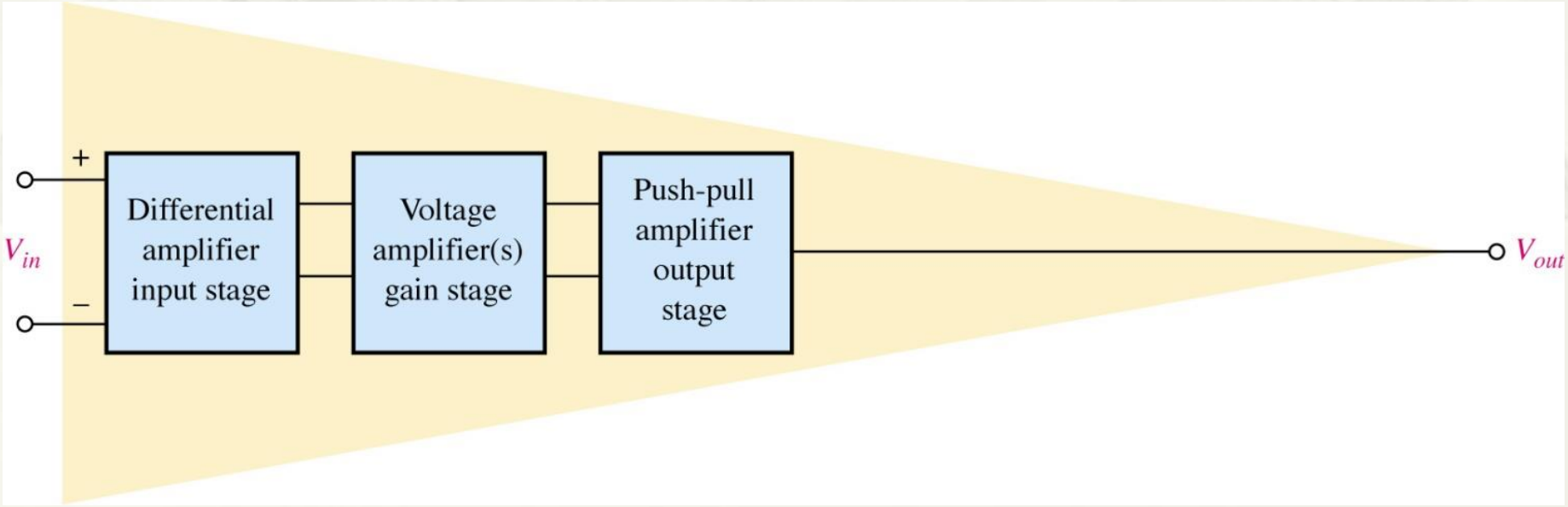
6-1 Introduction to Operational Amplifiers

- **Composition of operational amplifier**

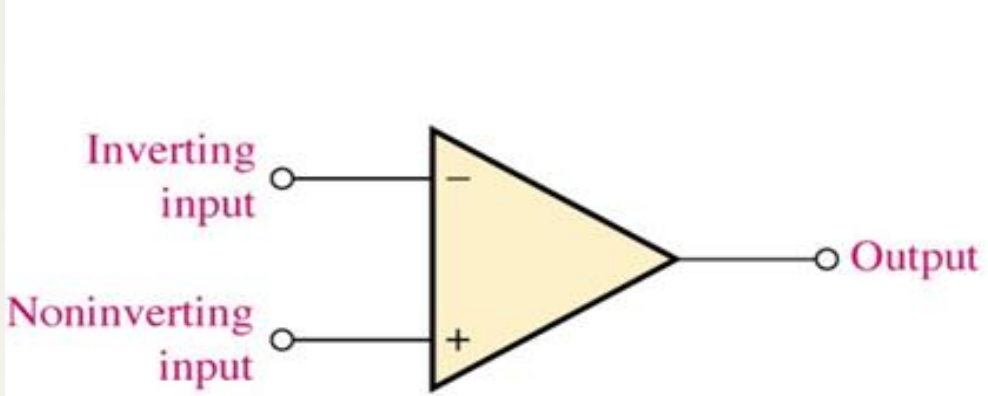


6-1 Introduction to Operational Amplifiers

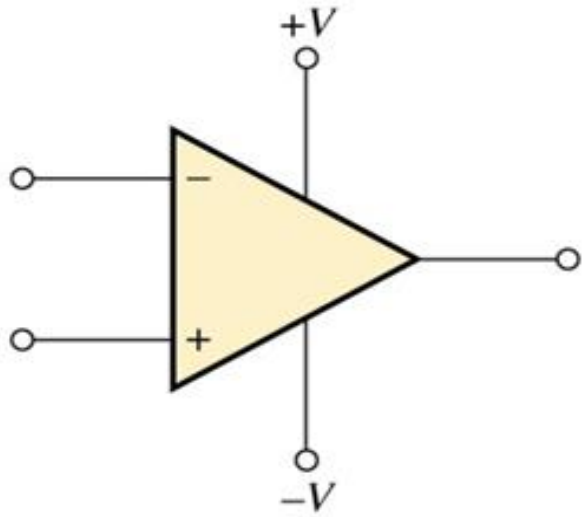
- **Composition of operational amplifier**
  - Input stage: differential amplifier
  - Gain stage: class A voltage amplifier
  - Output stage: push-pull class B amplifier



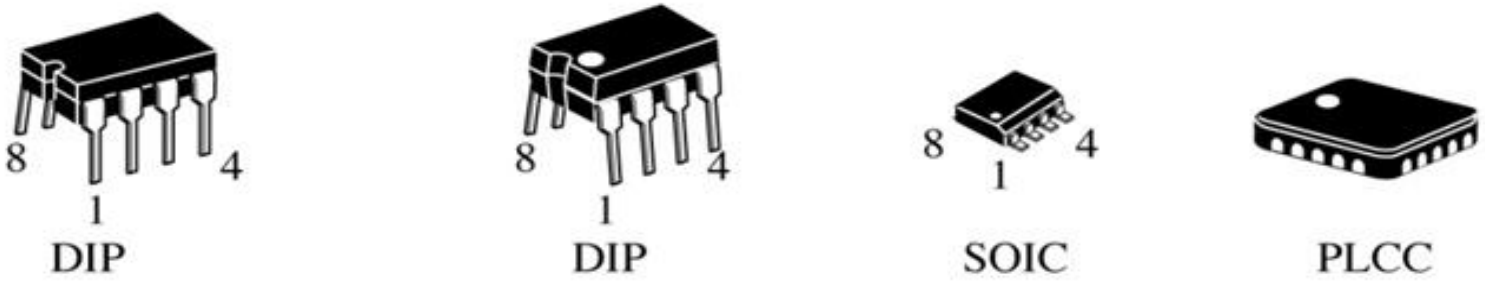
Symbol and Terminals



(a) Symbol



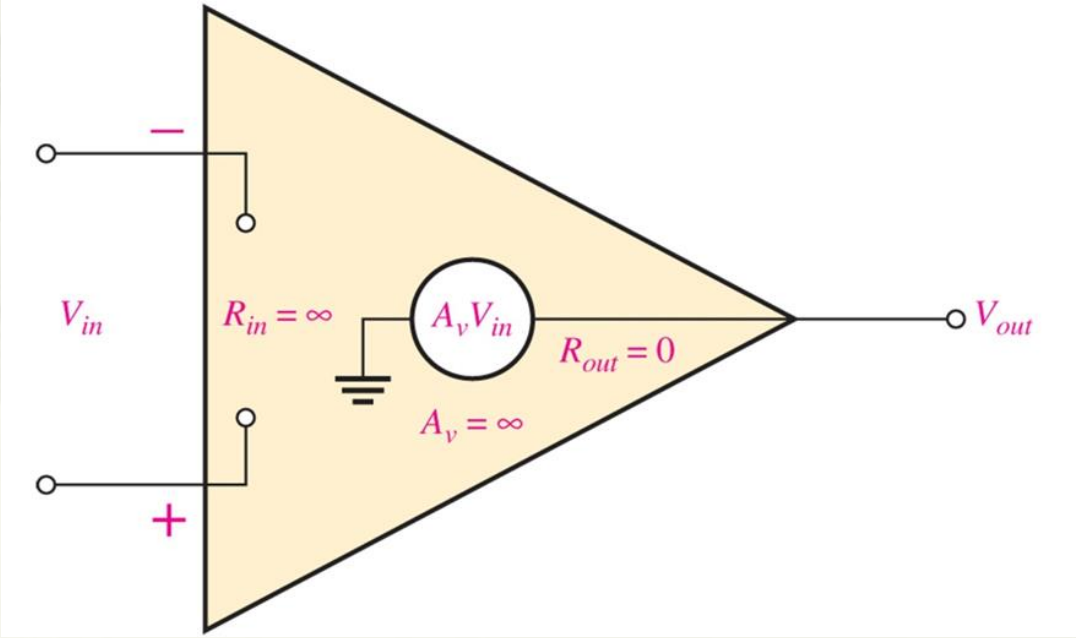
(b) Symbol with dc supply connections



(c) Typical packages. Looking from the top, pin 1 always is to the left of the notch or dot on the DIP and SOIC packages. The dot indicates pin 1 on the plastic-leaded chip carrier (PLCC) package.

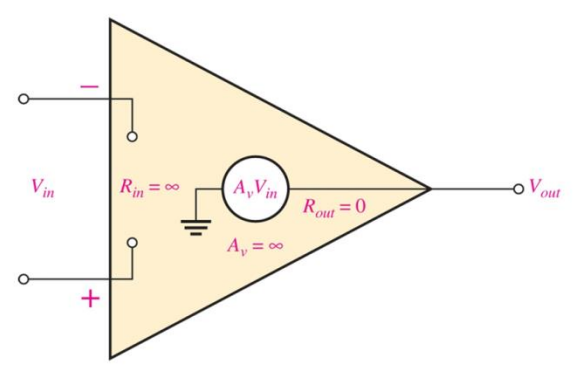
The Ideal Op-Amp

- **The ideal op-amp**
  - Infinite voltage gain
  - Infinite input resistance (open)
  - Zero output resistance



The Ideal Op-Amp

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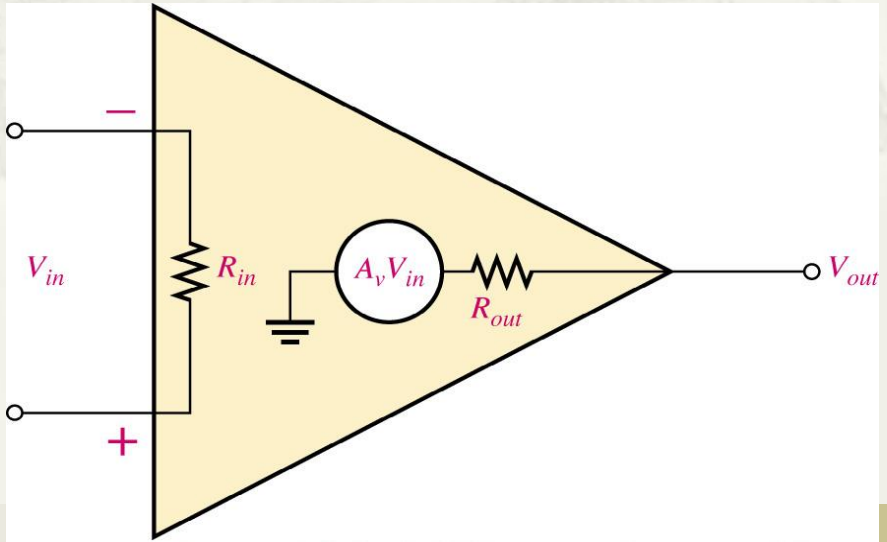


For Linear application

- **Another description of the ideal op-amp**
  - Virtual short: $v_+ = v_-$
  - Virtual open: $i_+ = i_- = 0$

The Practical Op-Amp

- **The limitations of practical op-amp**
  - Peak-to-peak output voltage is usually limited to slightly less than the difference between the two supply voltage. ***Special case**: rail-to-rail op-amp*
  - Output current is limited by internal restrictions such as power dissipation and component ratings.
- **Characteristics**
  - High voltage gain
  - High input resistance
  - Low output resistance

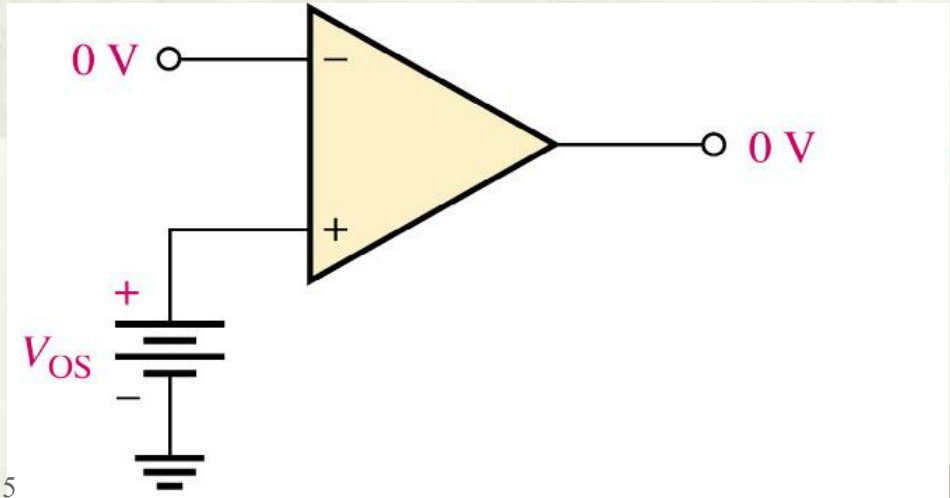


6-2 Op-Amp parameters

- Input Offset Voltage
- Input Bias Current
- Input Resistance
- Output Resistance
- Common-Mode Input Voltage Range
- Open-Loop Voltage Gain
- Common-Mode Rejection Ratio for an Op-Amp
- Slew Rate
- Frequency Response
- Comparison of Op-Amp Parameters

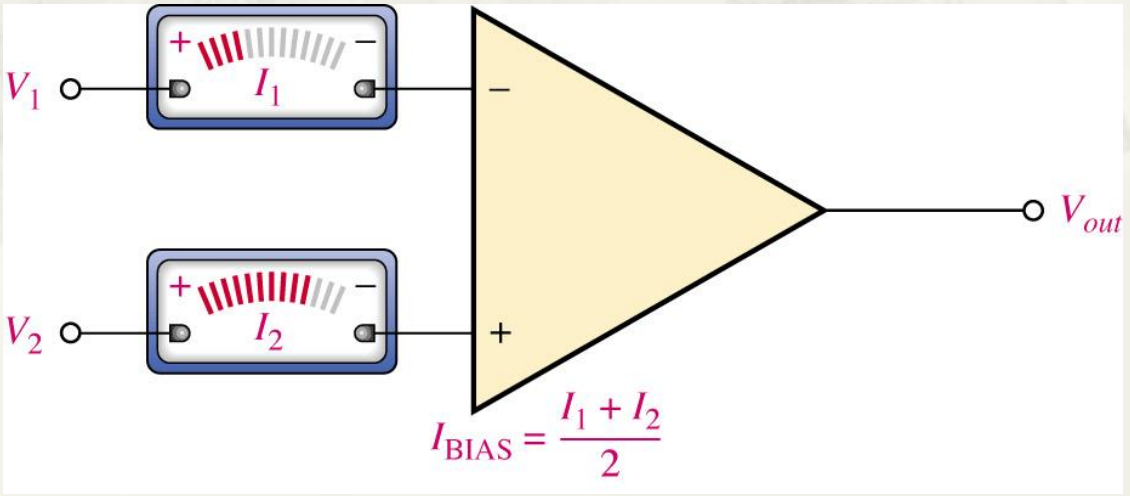
Input Offset Voltage

- **Origin**
  - The ideal op-amp produces zero volts for zero volts in.
  - In a practical op-amp, a small dc voltage, $V_{OUT(error)}$, appears at the output when no differential input voltage is applied.
- **Definition**
  - The input offset voltage (V_{OS}) is the differential dc voltage required between the inputs to force the output to zero volts.
- **Typical value**
  - In the range of several mV or less.
- **Ideal value**
  - 0 V



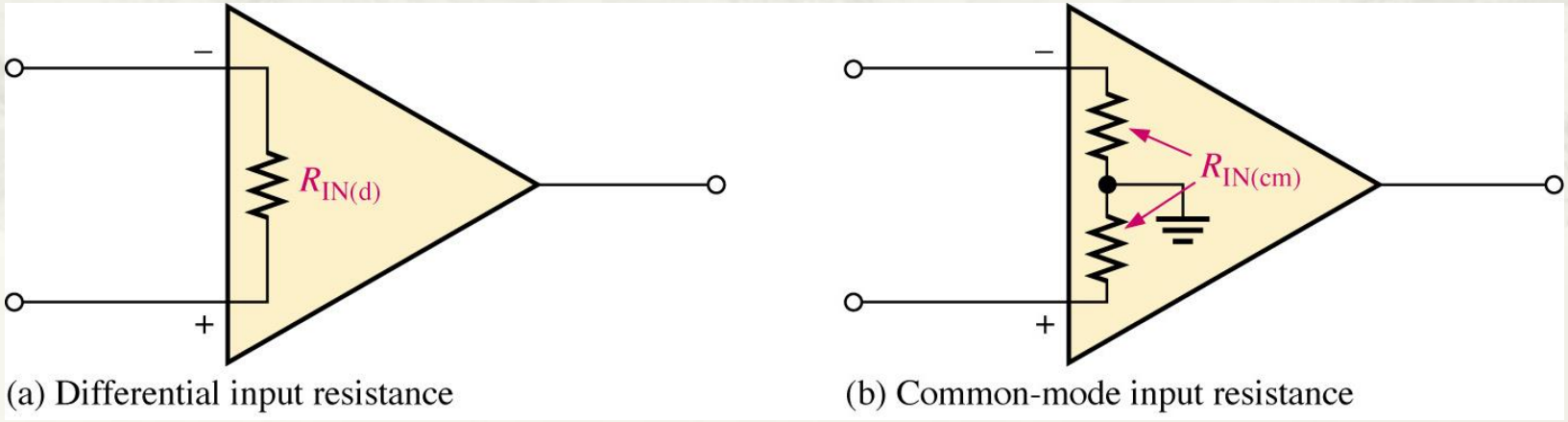
Input Bias Current

- **Definition**
  - The input bias current is the dc current required by the inputs of the amplifier to properly operate the first stage.
  - The input bias current is the average of both input currents.
- **Typical value**
  - 10-1000 nA
- **Ideal value**
  - 0 nA



Input Resistance

- **Differential input resistance**
  - The total resistance between the inverting and the noninverting inputs.
- **Common-mode input resistance**
  - The resistance between each input and ground.



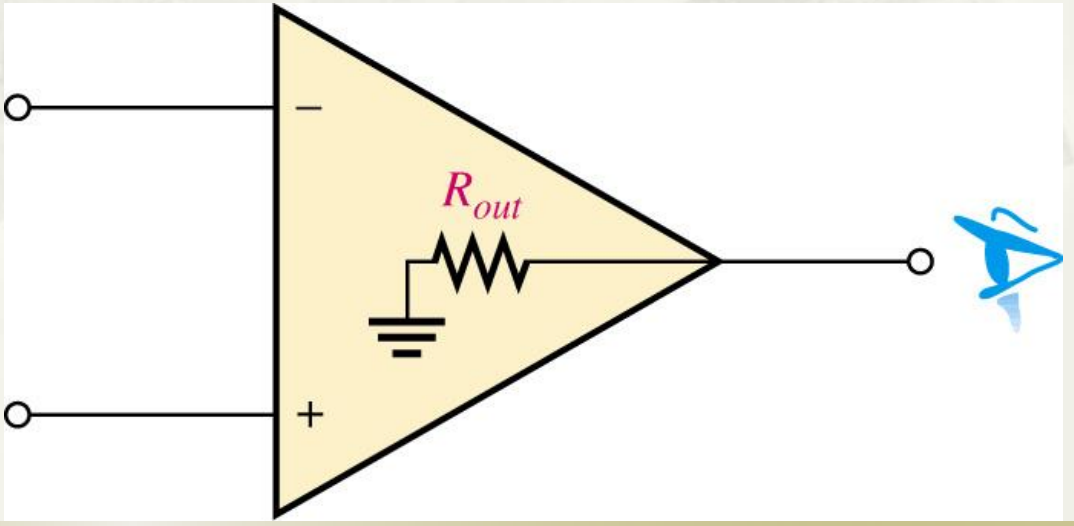
(a) Differential input resistance

(b) Common-mode input resistance

The input resistance is always very high, which is descripted in MΩ.

Output Resistance

- **Definition**
  - Op-amp output resistance is the resistance viewed from the output terminal of the op-amp.
- **Typical value**
  - From several ohms to tens of ohms
- **Ideal value**
  - 0 Ω

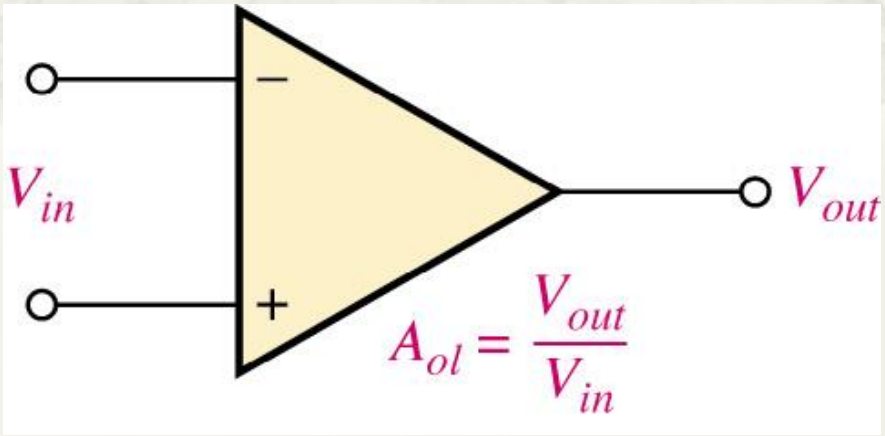


Common-Mode Input Voltage Range

- **Definition**
  - Common-mode input voltage range is the range of input voltages which, when applied to both inputs, will not cause clipping or other output distortion.
- **Typical value**
  - For general op-amp: 1 to 5 volts less than dc supply voltage
  - For rail-to-rail op-amp: as high as the dc supply voltage
- **Ideal value**
  - Infinite

Open-Loop Voltage Gain

- **Definition**
  - Open-loop voltage gain, A_{ol}(large-signal voltage gain) is the internal voltage gain of the device measured in the differential mode.
  - It represent the ratio of output voltage to input voltage when there are no external components.
- **Typical value**
  - About 1-1000 V/mV
- **Ideal value**
  - Infinite

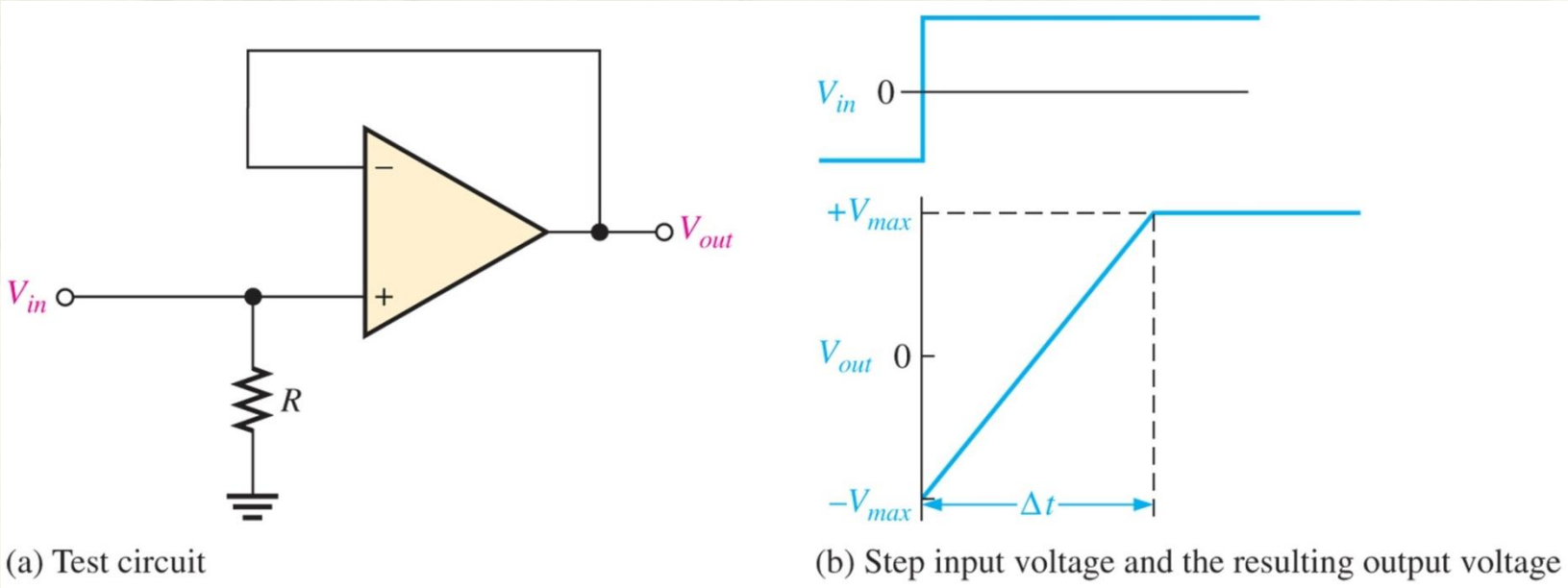


Common-Mode Rejection Ratio

- **Definition**
  - Common-Mode Rejection Ratio (CMRR) is the open-loop voltage gain A_{ol} divided by the common-mode gain.
  - $\mathrm{CMRR} = \dfrac{A_{ol}}{A_{cm}}$ **Expressed as a logarithmic ratio in decibels (dB)**
- **Physical meaning**
  - CMRR is a measure of an op-amp's ability to reject common-mode signals.
- **Typical value**
  - About 10-100 dB
- **Ideal value**
  - Infinite

Slew Rate

- **Definition**
  - The maximum rate of change of the output voltage in response to a step input voltage is the slew rate of an op-amp.
  - Slew rate $= \frac{\Delta V_{out}}{\Delta t}$, where $\Delta V_{out} = +V_{max} - (-V_{max})$



(a) Test circuit

(b) Step input voltage and the resulting output voltage

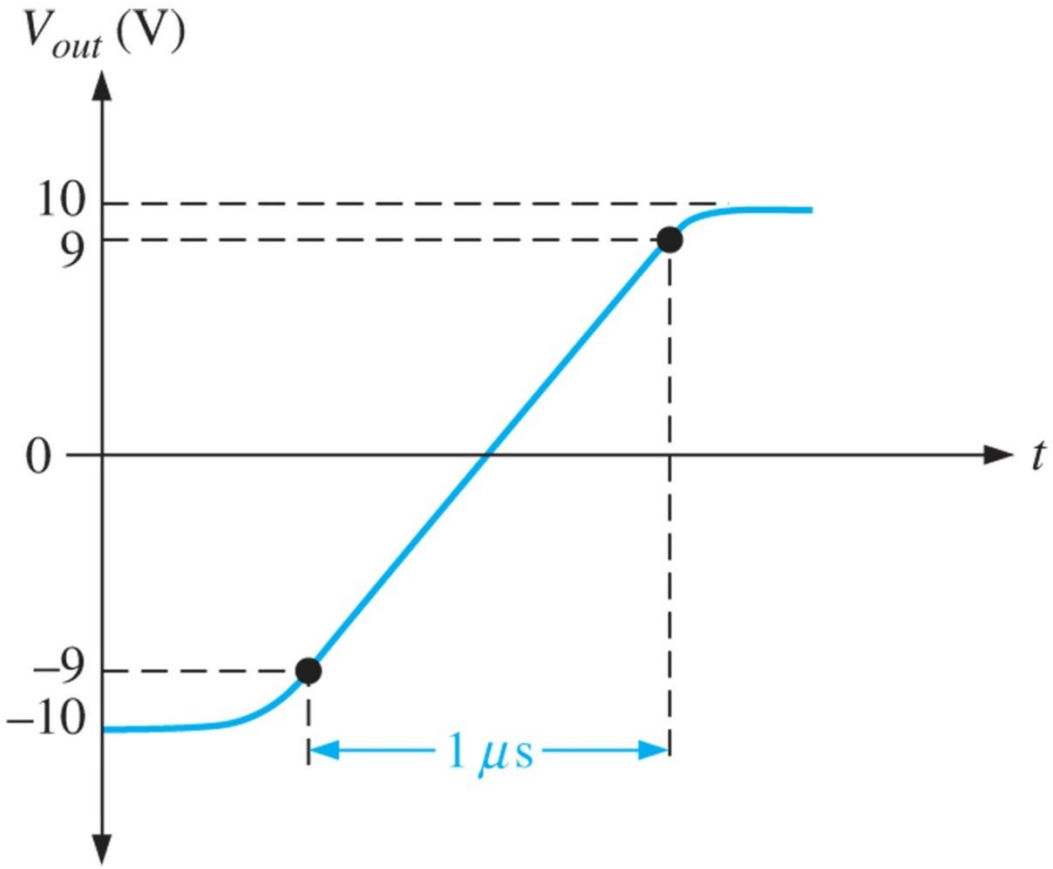
Slew Rate

- **Physical meaning**
  - The slew rate is dependent upon the high-frequency response of the amplifier stages within the op-amp.
- **Typical value**
  - About 1-1000 V/μs
- **Ideal value**
  - Infinite

Example 6-2

Problem

The output voltage of a certain op-amp appears as shown in Figure 6–11 in response to a step input. Determine the slew rate.



Example 6-2

Solution

The output goes from the lower to the upper limit in 1 μs. Since this response is not ideal, the limits are taken at the 90% points, as indicated, So, the upper limit is +9 V and the lower limit is −9 V. The slew rate is

$$\text{Slew rate} = \frac{\Delta V}{\Delta t} = \frac{+9\text{V} - (-9\text{V})}{1\ \mu\text{s}} = \mathbf{18\ V/\mu s}$$

Frequency Response

- The internal amplifier stages that make up an op-amp have voltage gain limited by junction capacitances.
- An op-amp has no internal coupling capacitors; therefore, the low frequency response extend down to dc (0Hz).

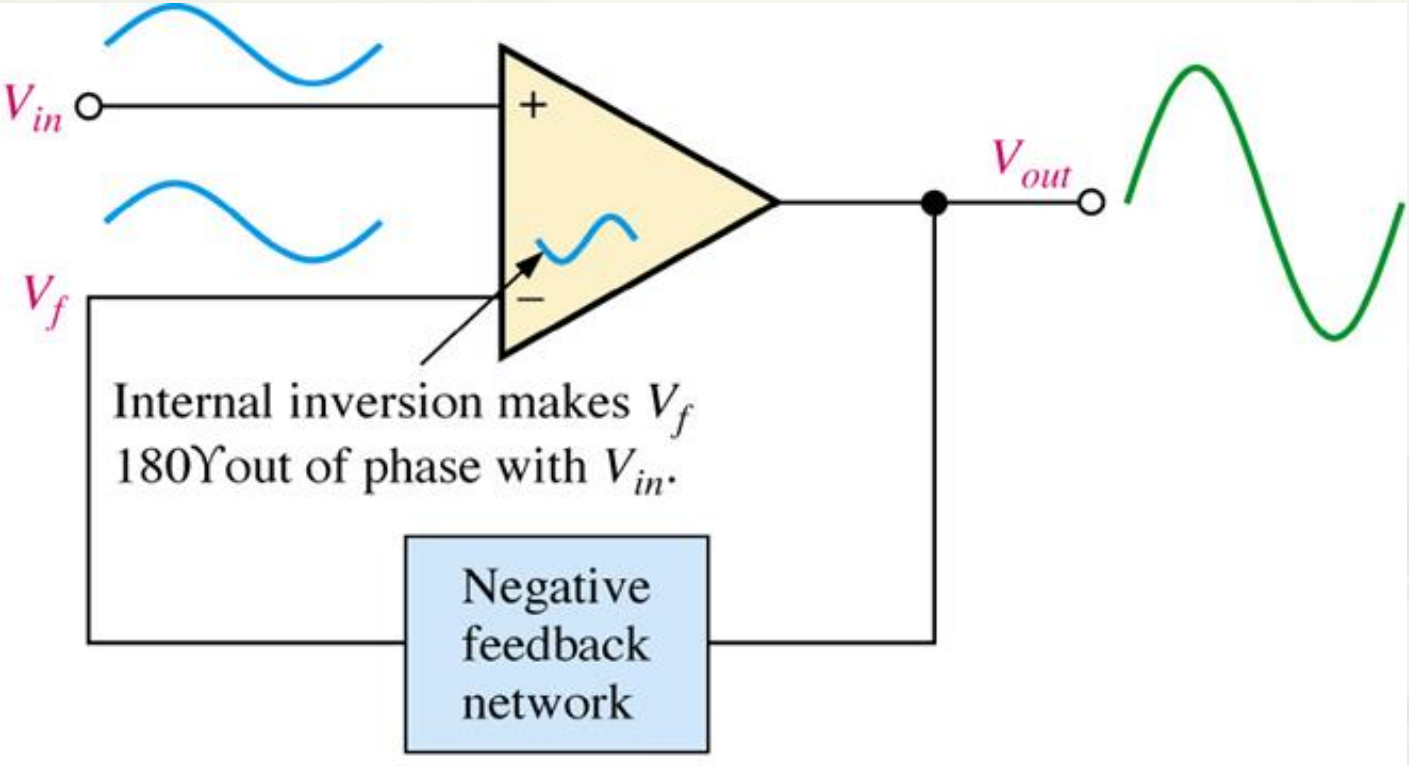
Comparison of Op-Amp Parameters

OP-AMP	Input offset voltage (mV) (max)	Input bias current (nA) (max)	Input resistance (MΩ) (min)	Open-loop gain (typ)	Slew rate (V/μs) (typ)	CMRR (dB) (min)	Comment
LM741C	6	500	0.3	200,000	0.5	70	Industry standard
LM101A	7.5	250	1.5	160,000	—	80	General-purpose
OP113E	0.075	600	—	2,400,000	1.2	100	Low noise, low drift
OP177A	0.01	1.5	26	12,000,000	0.3	130	Ultra precision
OP184E	0.065	350	—	240,000	2.4	60	Precision, rail-to-rail*
AD8009AR	5	150	—	—	5500	50	*BW* = 700 MHz, ultra fast, low distortion, current feedback
AD8041A	7	2000	0.16	56,000	160	74	*BW* = 160 MHz, rail-to-rail
AD8055A	5	1200	10	3500	1400	82	Very fast voltage feedback

Other Features

- **Most available op-amps have three important features**
  - Short-circuit protection
    - ✓ Keeps the circuit from being damaged if the output becomes shorted.
  - No latch-up
    - ✓ Prevents the op-amp from hanging up in one output state under certain input conditions.
  - Input offset nulling
    - ✓ Be achieved by an external potentiometer that sets the output voltage at precisely zero with zero input.

6-3 Negative Feedback in Op-Amps

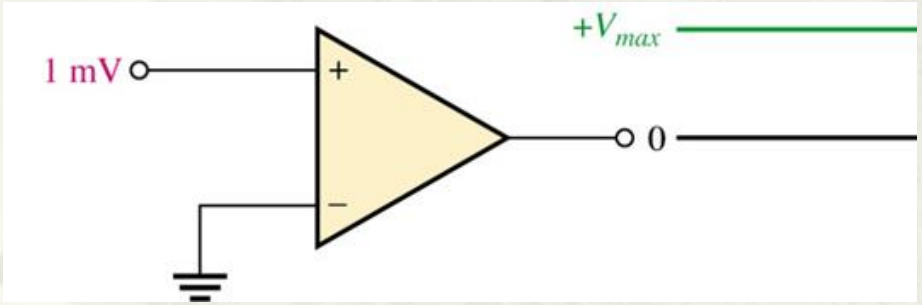


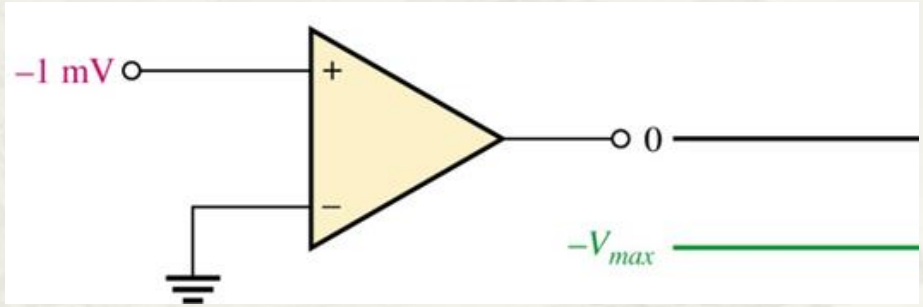
When negative feedback is present, the noninverting and inverting inputs are nearly identical.

Virtual short

6-3 Negative Feedback in Op-Amps

- Without negative feedback, an extremely small difference in the two input voltages drives the op-amp to its output limits and it becomes nonlinear.





6-3 Negative Feedback in Op-Amps

- The general effects of negative feedback on op-amp performance

	Voltage gain	Input resistance	Output resistance	Bandwidth
Without negative feedback	A_{ol} is too high for linear amplifier applications	Relatively high (see Table 6–1)	Relatively low	Relatively narrow (because the gain is so high)
With negative feedback	A_{cl} is set to desired value by the feedback network	Can be increased or reduced to a desired value depending on type of circuit	Can be reduced to a desired value	Significantly wider

6-4 Op-Amp Configurations

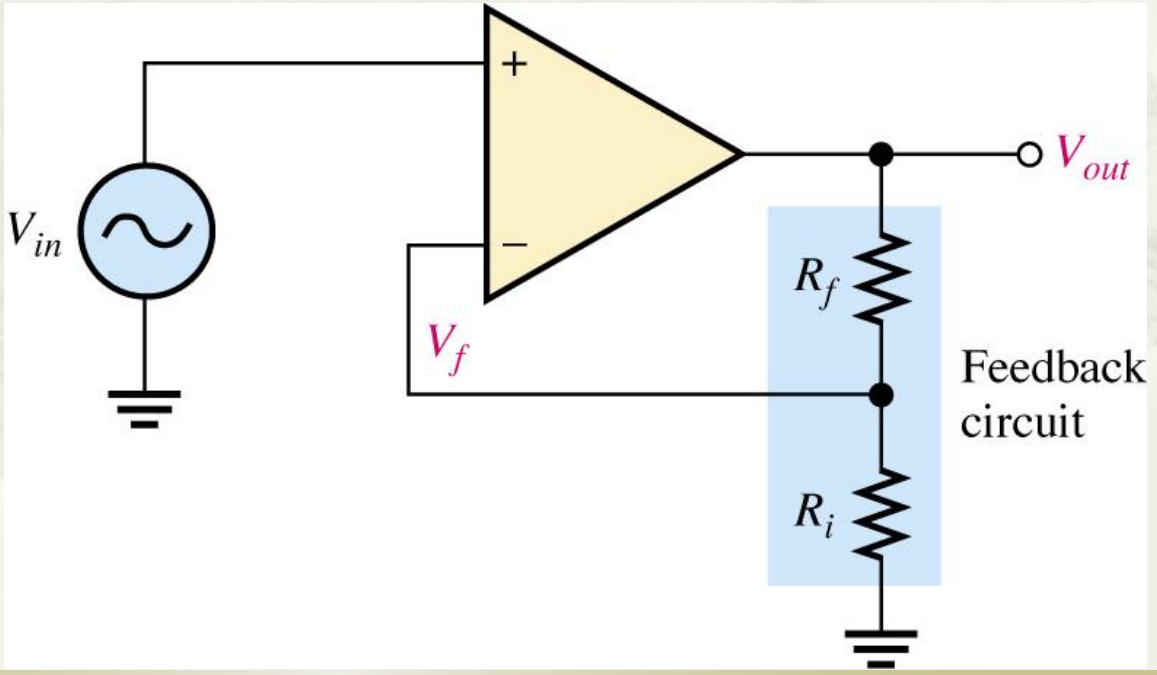
- Closed-Loop Voltage Gain
- Noninverting Amplifier
- Inverting Amplifier
- Effect of Negative Feedback on Bandwidth
- Gain-Bandwidth Product

Closed-Loop Voltage Gain

- **Closed-loop voltage gain**
  - The closed-loop voltage gain, A_{cl}, is the voltage gain of an op-amp with negative feedback.
  - The amplifier configuration consists of the op-amp and an external feedback circuit that connects the output to the inverting input.
  - The closed-loop voltage gain is then determined by the component values in the feedback circuit and can be precisely controlled by them.

Noninverting Amplifier

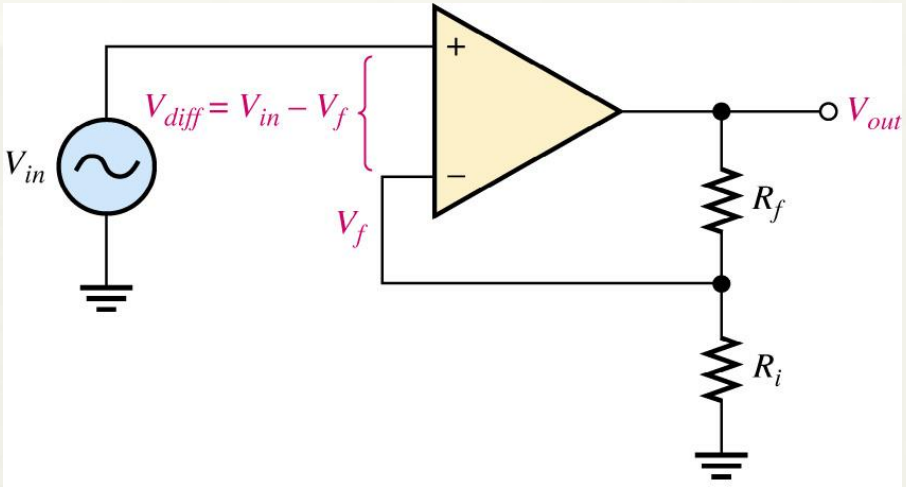
- **Noninverting amplifier**
  - An op-amp connected in a closed-loop configuration in which the input signal is applied to the noninverting input (+) is a noninverting amplifier.



Noninverting Amplifier

The differential voltage, V_{diff},

$$V_{diff} = V_{in} - V_f$$



As the result of the negative feedback and the high open-loop gain, A_{ol}. Therefore,

$$V_{diff} = 0\text{V} \quad \text{(Virtual short)}$$

$$V_{in} = V_f$$

$$V_f = \left(\frac{R_i}{R_i + R_f}\right) V_{out}$$

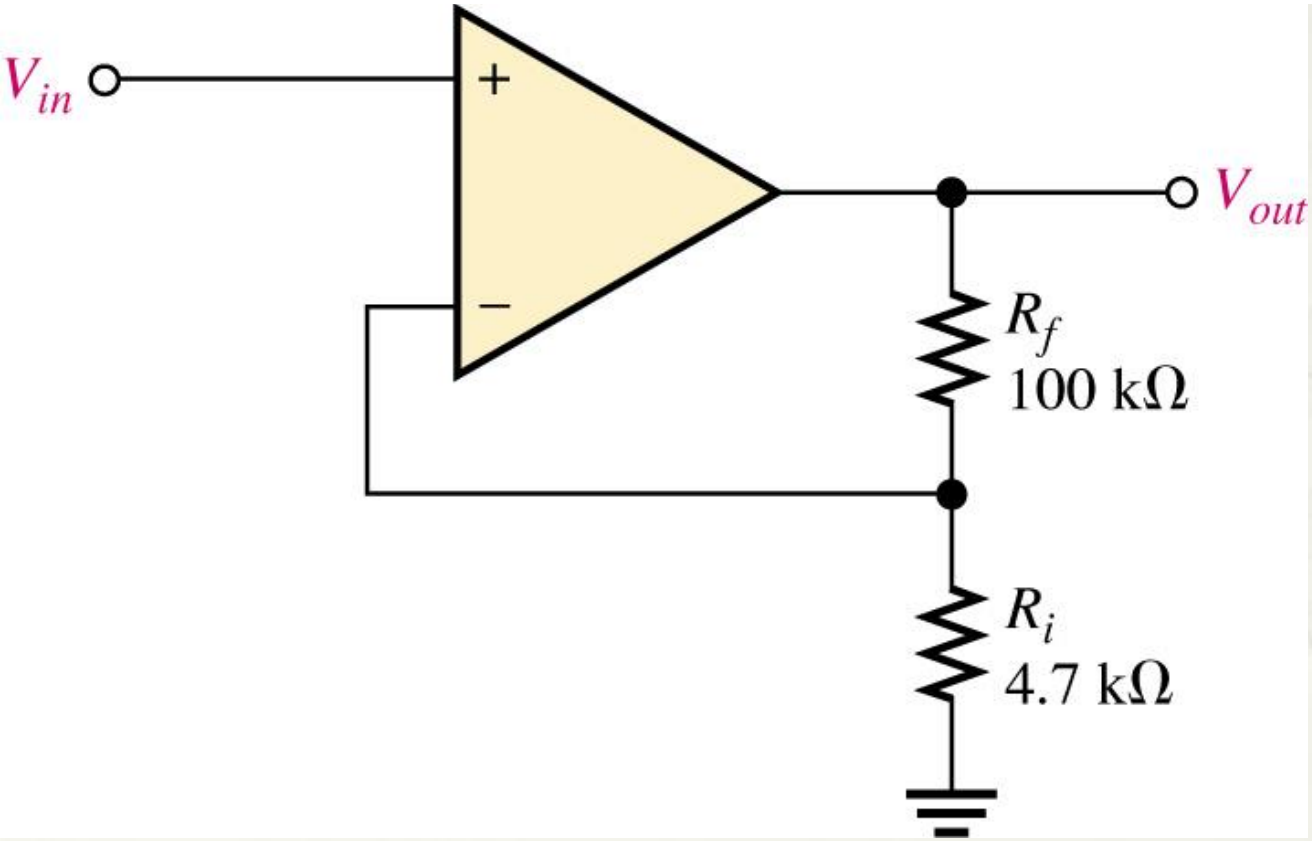
The closed-loop voltage gain of the noninverting (NI) amplifier:

$$A_{cl(NI)} = \frac{V_{out}}{V_{in}} = \left(\frac{R_i + R_f}{R_i}\right) = 1 + \frac{R_f}{R_i}$$

Example 6-3

Problem

Determine the closed-loop voltage gain of the amplifier in Figure 6–16.



Example 6-3

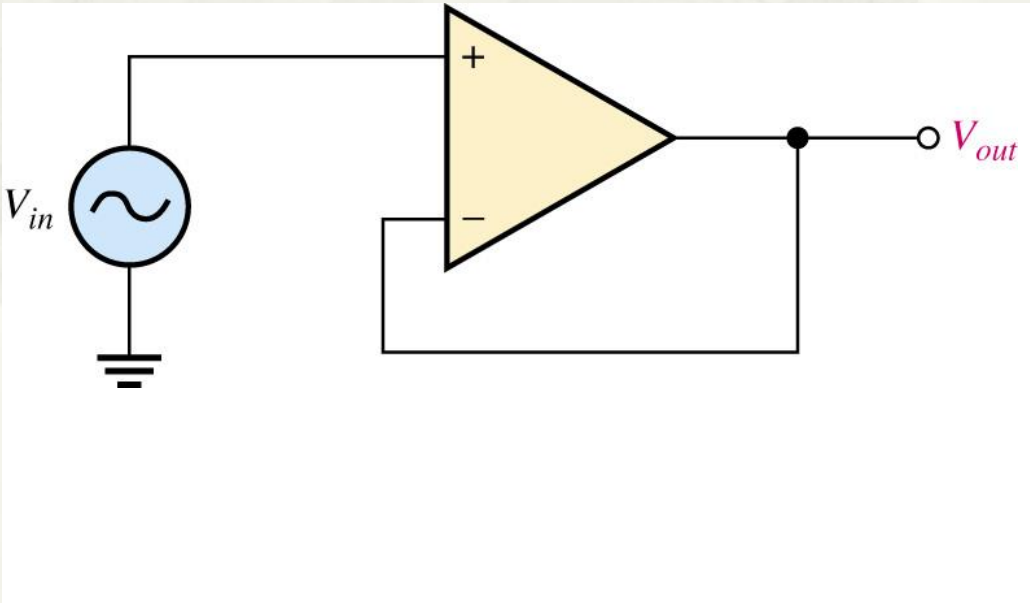
Solution

This is a noninverting op-amp. Therefore, the closed-loop voltage gain is

$$A_{cl(\mathrm{NI})} = \frac{R_f}{R_i} + 1 = \frac{100\ \mathrm{k\Omega}}{4.7\ \mathrm{k\Omega}} + 1 = \mathbf{22.3}$$

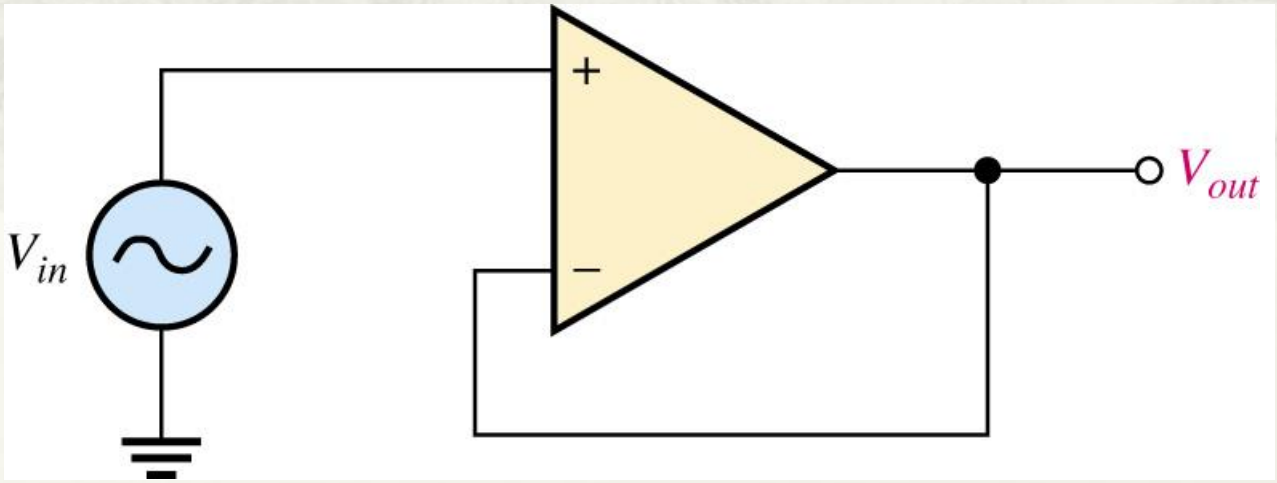
Noninverting Amplifier

- **Voltage-Follower**
  - The voltage-follower is a special case of the noninverting amplifier where all of the output voltage is fed back to the inverting input by a straight connection.
  - $A_{cl(\mathrm{VF})} = 1$



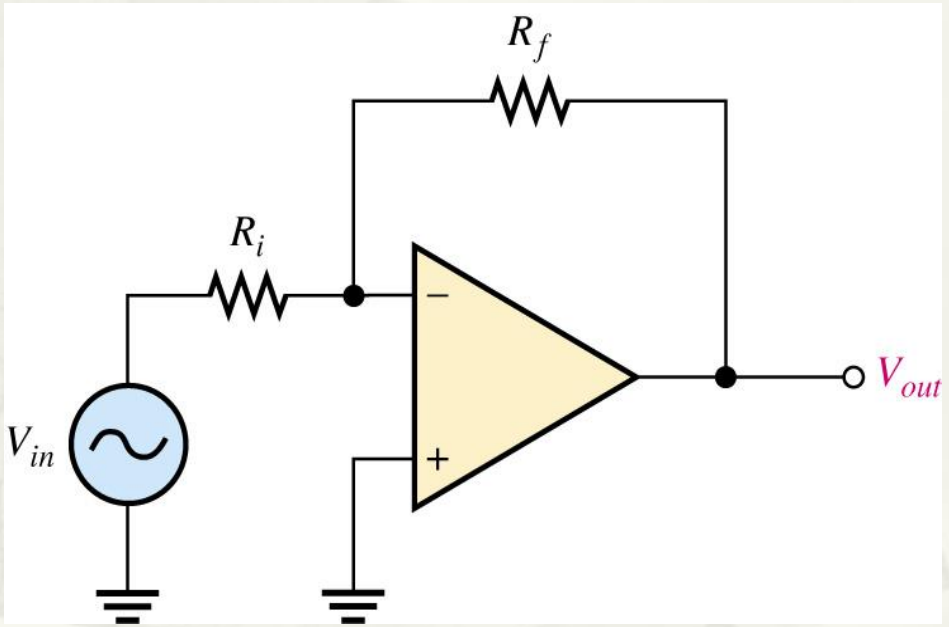
Noninverting Amplifier

- **Voltage-Follower**
  - The most important features of the voltage-follower configuration are its very high input resistance and its very low output resistance.
  - It is a nearly ideal buffer amplifier for interfacing high-resistance sources and low-resistance loads.

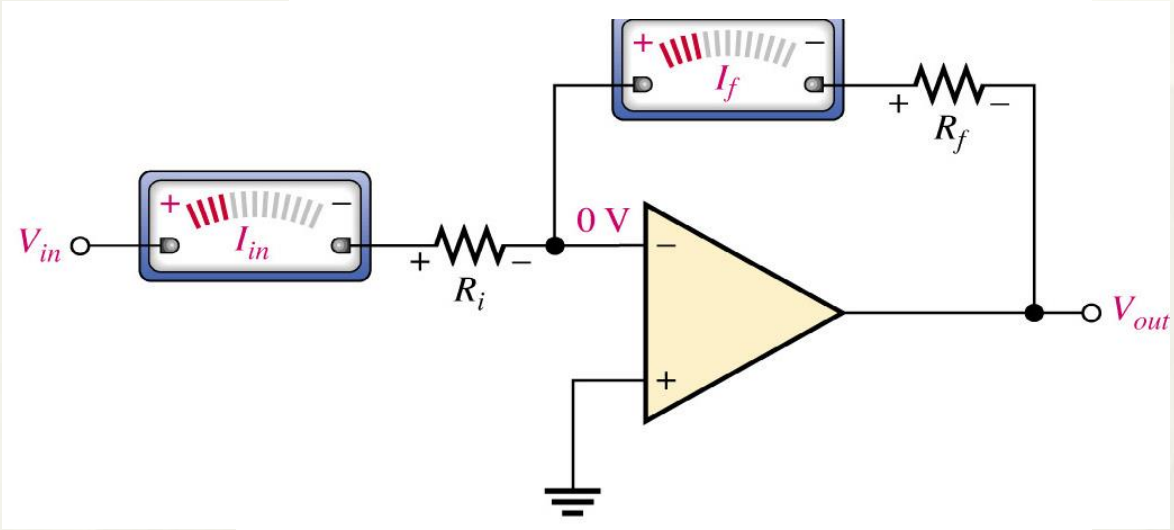


Inverting Amplifier

- **Inverting amplifier**
  - An op-amp connected in a closed-loop configuration in which the input signal is applied through a series resistor to the inverting input (-) is inverting amplifier.



Inverting Amplifier



Virtual ground

The closed-loop gain is independent of the op-amp's internal open-loop gain.

Virtual open ⇒ $I_f = I_{in}$

$$\frac{-V_{out}}{R_f} = \frac{V_{in}}{R_i} \Rightarrow \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_i}$$

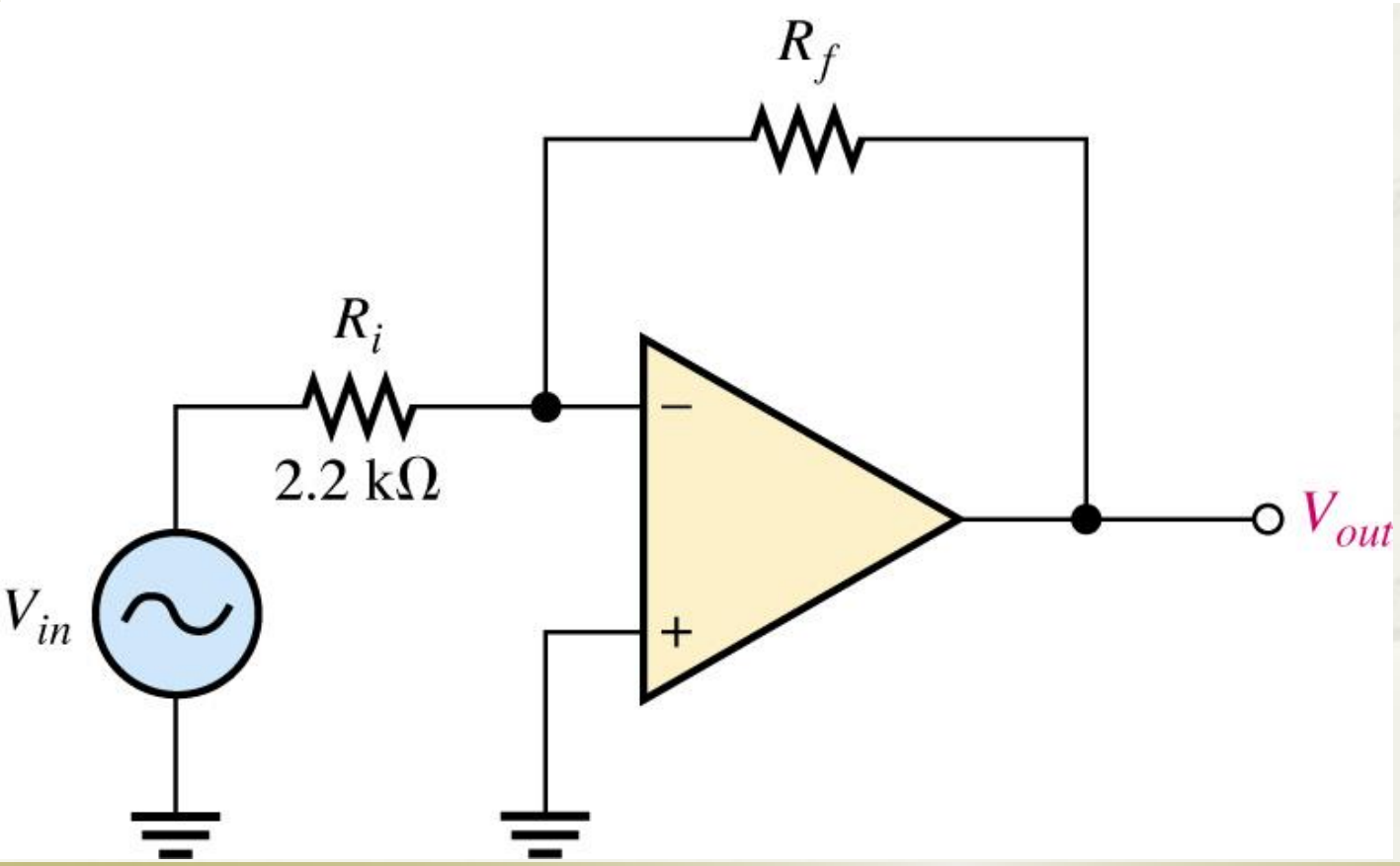
The closed-loop gain of the inverting amplifier:

$$A_{cl(I)} = \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_i}$$

Example 6-4

Problem

Given the op-amp configuration in Figure 6–20, determine the value of R_f required to produce a closed-loop voltage gain of -100.



Example 6-4

Solution

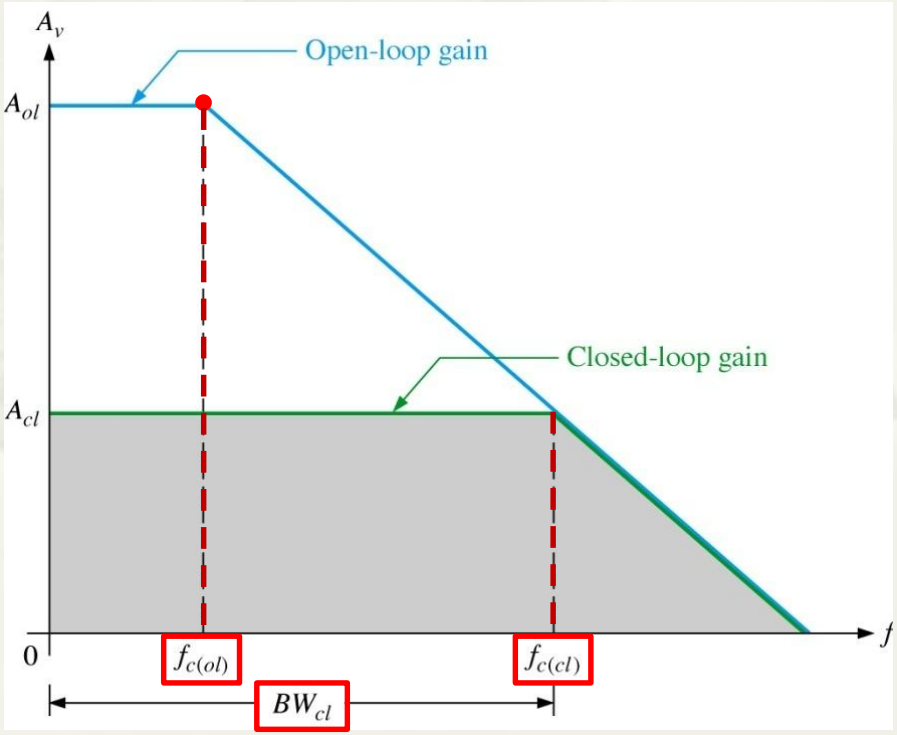
Knowing that $R_i = 2.2\ \text{k}\Omega$ and $A_{cl(\text{I})} = -100$, calculate R_f as follows:

$$A_{cl(\text{I})} = -\frac{R_f}{R_i}$$

$$R_f = -A_{cl(\text{I})}R_i = -(-100)(2.2\ \text{k}\Omega) = \mathbf{220\ k\Omega}$$

Effect of Negative Feedback on Bandwidth

- When the open-loop gain of an op-amp is reduced by negative feedback, the bandwidth is increased.
- Beyond the closed-loop critical frequency, the closed-loop gain decreases at the same rate (called the roll-off rate) as the open-loop gain.



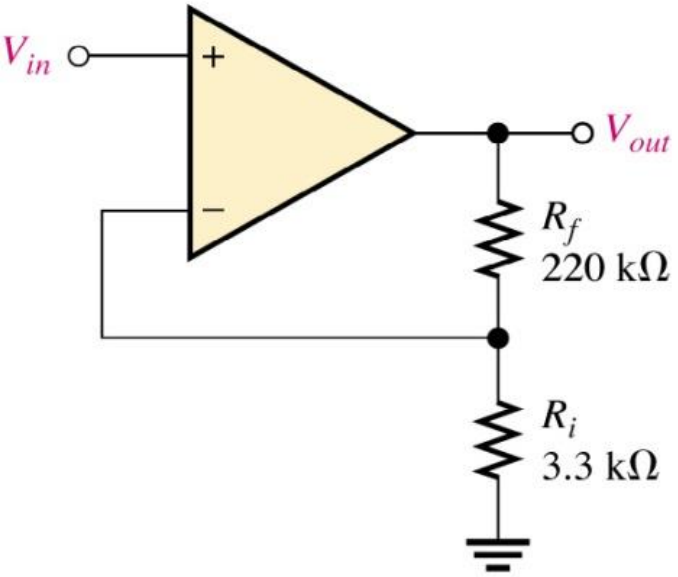
Gain-Bandwidth Product

- An increase in closed-loop gain cause a decrease in the bandwidth and vice versa, such that *the product of gain and bandwidth is a constant*, as long as the roll-off rate is a fixed -20dB/decade.
  - $A_{cl}f_{c(cl)} = A_{ol}f_{c(ol)}$
- The gain-bandwidth product is always equal to the frequency at which the op-amp's open-loop gain is unity (unity-bandwidth).
  - $A_{cl}f_{c(cl)} = \text{unit} - \text{gain bandwidth}$

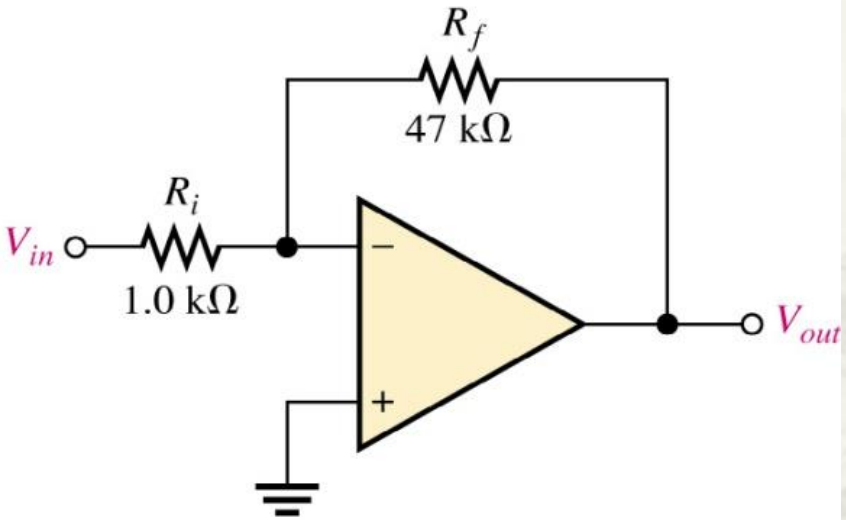
Example 6-5

Problem

Determine the bandwidth of each of the amplifiers in Figure 6–22. Both op-amps have a unity-gain bandwidth of 3 MHz.



(a)



(b)

Example 6-5

Solution

(a) For the noninverting amplifier in Figure 6–22(a), the closed-loop gain is

$$A_{cl(\mathrm{NI})} = \frac{R_f}{R_i} + 1 = \frac{220\ \mathrm{k\Omega}}{3.3\ \mathrm{k\Omega}} + 1 = 67.7$$

Use Equation 6–6 and solve for $f_{c(cl)}$ (where $f_{c(cl)} = BW_{cl}$).

$$f_{c(cl)} = BW_{cl} = \frac{\text{unity-gain } BW}{A_{cl}}$$

$$BW_{cl} = \frac{3\ \mathrm{MHz}}{67.7} = \mathbf{44.3\ kHz}$$

Example 6-5

Solution

(b) For the inverting amplifier in Figure 6--22(b), the closed-loop gain is

$$A_{cl(\mathrm{I})} = -\frac{R_f}{R_i} = -\frac{47\ \mathrm{k\Omega}}{1.0\ \mathrm{k\Omega}} = -47$$

Using the absolute value of $A_{cl(\mathrm{I})}$, the closed-loop bandwidth is

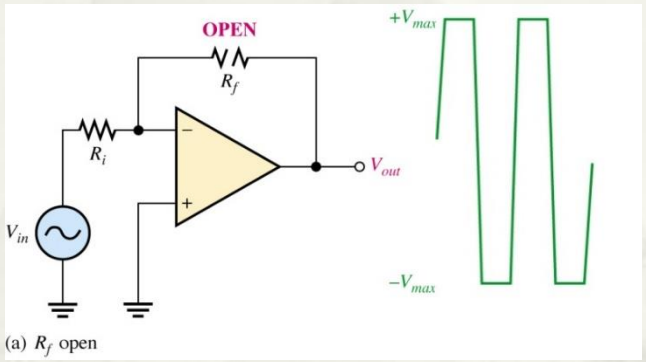
$$BW_{cl} = \frac{3\ \mathrm{MHz}}{47} = \mathbf{63.8\ kHz}$$

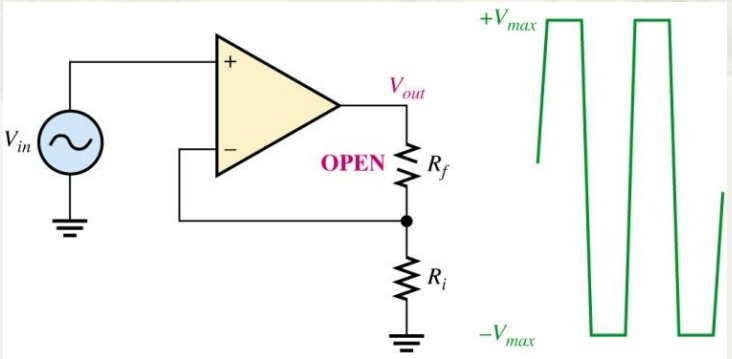
6-5 Troubleshooting

- Faults in the Noninverting Amplifier
- Faults in the Voltage-Follower
- Faults in the Inverting Amplifier

Faults in the Noninverting Amplifier

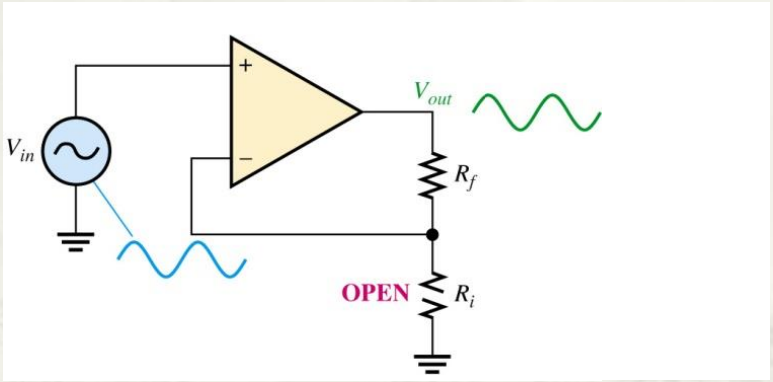
- **Power supply check**
  - The positive and negative supply voltages should be measured on the op-amp's pins with respect to a nearby circuit ground.
- **Open R_f**

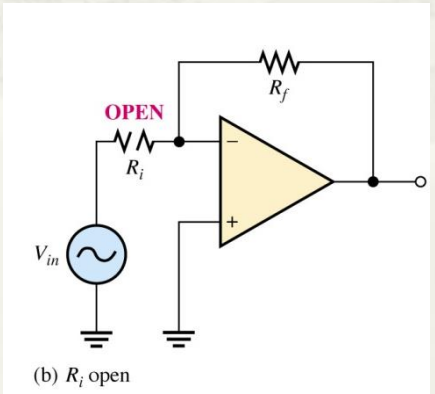




Faults in the Noninverting Amplifier

- **Power supply check**
  - The positive and negative supply voltages should be measured on the op-amp's pins with respect to a nearby circuit ground.
- **Open R_f**
- **Open R_i**



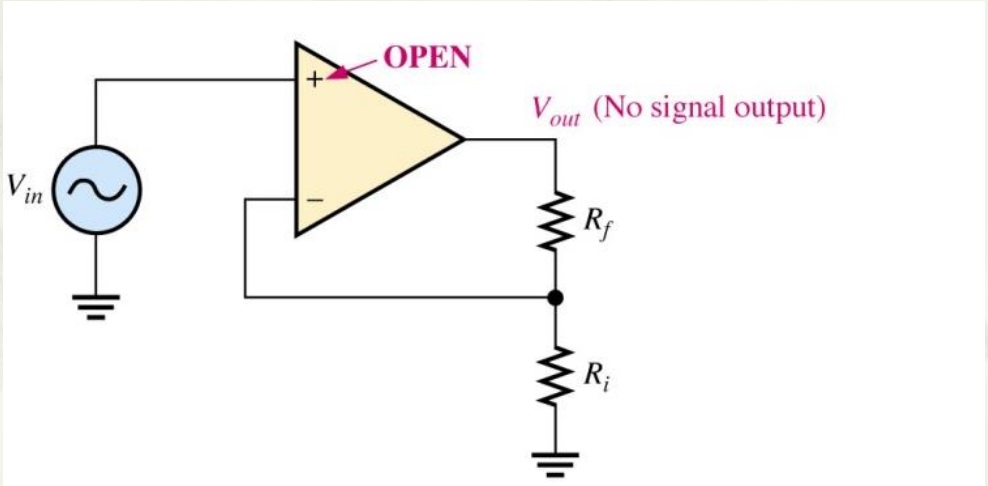


(b) R_i open

No ac signal out (output is saturated)

Faults in the Noninverting Amplifier

- **Power supply check**
  - The positive and negative supply voltages should be measured on the op-amp's pins with respect to a nearby circuit ground.
- **Open R_f**
- **Open R_i**
- **Internally Open Noninverting Op-Amp Input**
- **Other Op-Amp Faults**
- **Faults in the Voltage-Follower**



KEY OBJECTIVES

1. Describe the basic op-amp and its characteristics
2. Discuss several op-amp parameters
3. Explain negative feedback in op-amp circuits
4. Analyze three op-amp configurations and explain the closed-loop frequency response of an op-amp
5. Troubleshoot op-amp circuits

HOMEWORK

- **Chapter Checkup (all)**
  - Just do on your textbook
- **Questions (all)**
  - Odd-numbered in your mind, even-numbered on exercise book
- **Basic Problems**
  - 2, 3, 5, 8, 10
- **Basic-plus Problems**
  - 11, 18, 19, 21

RESEARCH PROJECT 3

- **Types of Operational Amplifiers**

  Write a report with respect to all of the following subjects:

  - Comparison of voltage feedback op-amp (VFB) and current feedback op-amp (CFB)
  - How to design an op-amp application circuit under single power supply condition
  - Features, model selection, applications and typical model of special purpose op-amps, including instrument amplifier, isolation amplifier, low-noise amplifier, wide-bandwidth amplifier, high speed amplifier, rail-to-rail amplifier, logarithm amplifier, etc.